1. Core definitions and notation

Correlation sequences

$$r_{xx}(k) = \mathrm{E}[x(n)x^*(n-k)]\,, \qquad r_{xy}(k) = \mathrm{E}[x(n)y^*(n-k)]$$

Bilateral z-transforms

$$\Phi_{xx}(z) = \sum_{k=-\infty}^{\infty} r_{xx}(k)z^{-k}, \qquad \Phi_{xy}(z) = \sum_{k=-\infty}^{\infty} r_{xy}(k)z^{-k}$$

Symmetry identities

$$r_{yx}(k) = r_{xy}^*(-k), \qquad \Phi_{yx}(z) = \Phi_{xy}^*(1/z^*)$$

White noise $v(n)$ with variance σ_v^2

$$r_{vv}(k) = \sigma_v^2\,\delta(k), \qquad \Phi_{vv}(z) = \sigma_v^2$$

LTI system

$$y(n) = h(n) * x(n), \qquad H(z) = \sum_n h(n)z^{-n}$$

For real-valued problems, replace $(\cdot)^{\mathsf{H}}$ by $(\cdot)^{\mathsf{T}}$ and conjugates by ordinary transposes.

2. LTI relations in the z-domain

If $y(n)$ is the output of $H(z)$ driven by $x(n)$,

$$\Phi_{yx}(z) = H(z)\Phi_{xx}(z)$$
$$\Phi_{xy}(z) = H^*(1/z^*)\,\Phi_{xx}(z)$$
$$\Phi_{yy}(z) = H(z)H^*(1/z^*)\,\Phi_{xx}(z)$$

If $d(n)$ is a third process,

$$\Phi_{yd}(z) = H(z)\Phi_{xd}(z), \qquad \Phi_{dy}(z) = H^*(1/z^*)\,\Phi_{dx}(z)$$

Time-domain cross-correlation form:

$$r_{yx}(k) = \sum_m h(m)\,r_{xx}(k-m)$$

3. White noise passed through $H(z)$

For white-noise input $v(n)$ and output $u(n)$,

$$\Phi_{vu}(z) = \sigma_v^2 H^*(1/z^*)$$
$$\Phi_{uu}(z) = H(z)H^*(1/z^*)\,\Phi_{vv}(z) = \sigma_v^2 H(z)H^*(1/z^*)$$
$$r_{vu}(k) = \sigma_v^2 h^*(-k)$$
$$r_{uu}(k) = \sigma_v^2 \sum_n h(n)h^*(n-k)$$

For FIR $H(z) = \sum_{n=0}^{N-1} h(n)z^{-n}$,

$$\Phi_{uu}(z) = \sigma_v^2 \sum_{k=-(N-1)}^{N-1} \alpha_h(k)z^{-k}$$

$$\alpha_h(k) = \sum_{m=\max(0,-k)}^{\min(N-1,N-1-k)} h(m+k)h^*(m)$$

Special case:

$$H(z) = \frac{1}{1-az^{-1}}, \quad |a| < 1$$

$$\Phi_{vu}(z) = \frac{\sigma_v^2}{1-a^*z}, \qquad \Phi_{uu}(z) = \frac{\sigma_v^2}{(1-az^{-1})(1-a^*z)}$$

4. Random-phase sinusoid plus white noise

If

$$x(n) = v(n) + \sin(\omega_0 n + \theta), \qquad \theta \sim U[-\pi, \pi]$$

then

$$\mathrm{E}[\sin(\omega_0 n + \theta)\sin(\omega_0(n-k) + \theta)] = \tfrac{1}{2}\cos(\omega_0 k)$$

so

$$r_{xx}(k) = \sigma_v^2\delta(k) + \tfrac{1}{2}\cos(\omega_0 k)$$

$$\Phi_{xx}(z) = \sigma_v^2 + \tfrac{1}{2}\sum_{k=-\infty}^{\infty}\cos(\omega_0 k)z^{-k}$$

Useful decomposition:

$$\tfrac{1}{2}\cos(\omega_0 k) = \tfrac{1}{4}e^{j\omega_0 k} + \tfrac{1}{4}e^{-j\omega_0 k}$$

Because $\cos(\omega_0 k)$ is two-sided and nondecaying, the bilateral z-transform has **no ROC**.

4A. Trig identities and phase averaging

Product-to-sum identities:

$$\sin A\sin B = \tfrac{1}{2}\cos(A-B) - \tfrac{1}{2}\cos(A+B)$$
$$\cos A\cos B = \tfrac{1}{2}\cos(A-B) + \tfrac{1}{2}\cos(A+B)$$
$$\sin A\cos B = \tfrac{1}{2}\sin(A+B) + \tfrac{1}{2}\sin(A-B)$$

Euler forms:

$$\cos\alpha = \tfrac{1}{2}\left(e^{j\alpha} + e^{-j\alpha}\right), \qquad \sin\alpha = \tfrac{1}{2j}\left(e^{j\alpha} - e^{-j\alpha}\right)$$

Phasor / angle-form reminder:

$$A\cos(\omega t + \theta) = \tfrac{1}{2}(A\angle\theta)e^{j\omega t} + \tfrac{1}{2}(A\angle-\theta)e^{-j\omega t}$$

Unit-amplitude special case:

$$\cos(\omega t + \theta) = \tfrac{1}{2}(1\angle\theta)e^{j\omega t} + \tfrac{1}{2}(1\angle-\theta)e^{-j\omega t}$$

If $\theta \sim U[-\pi, \pi]$, then averaging over one full period gives

$$\mathrm{E}_\theta[\sin(c+\theta)] = 0, \quad \mathrm{E}_\theta[\cos(c+\theta)] = 0$$
$$\mathrm{E}_\theta[\sin(c+2\theta)] = 0, \quad \mathrm{E}_\theta[\cos(c+2\theta)] = 0$$

Useful reduction:

$$\sin(\omega_0 n + \theta)\sin(\omega_0(n-k) + \theta) = \tfrac{1}{2}\cos(\omega_0 k) - \tfrac{1}{2}\cos(2\omega_0 n - \omega_0 k + 2\theta)$$

so after expectation over θ,

$$\mathrm{E}_\theta[\sin(\omega_0 n + \theta)\sin(\omega_0(n-k) + \theta)] = \tfrac{1}{2}\cos(\omega_0 k)$$

5. Additive-noise observation model

If

$$x(n) = d(n) + v(n)$$

with $v(n)$ zero mean and uncorrelated with $d(n)$, then

$$p_{dx}(m) = \mathrm{E}[d(n)x(n-m)] = r_d(m)$$
$$r_x(m) = \mathrm{E}[x(n)x(n-m)] = r_d(m) + r_v(m)$$

Matrix Wiener-Hopf form:

$$(\mathbf{R}_d + \mathbf{R}_v)\mathbf{w}_o = \mathbf{p}_{dx} = \mathbf{p}_d$$

If the noise is white,

$$\mathbf{R}_v = \sigma_v^2\mathbf{I}$$

6. Wiener filter setup

For an M-tap FIR Wiener filter,

$$\mathbf{x}(n) = \begin{bmatrix} x(n) & x(n-1) & \cdots & x(n-M+1) \end{bmatrix}^T$$

$$y(n) = \mathbf{w}^{\mathsf{H}}\mathbf{x}(n), \qquad e(n) = d(n) - y(n)$$

Statistics:

$$\mathbf{R} = \mathrm{E}[\mathbf{x}(n)\mathbf{x}^{\mathsf{H}}(n)], \qquad \mathbf{p} = \mathrm{E}[d^*(n)\mathbf{x}(n)]$$

Toeplitz autocorrelation matrix and cross-correlation vector:

$$\mathbf{R}_{ij} = r_x(i-j), \qquad p_i = p_{dx}(i), \qquad i,j = 0,\ldots,M-1$$

Complex performance function:

$$\xi(\mathbf{w}) = \mathrm{E}[|e(n)|^2] = \mathrm{E}[|d(n)|^2] - \mathbf{w}^{\mathsf{H}}\mathbf{p} - \mathbf{p}^{\mathsf{H}}\mathbf{w} + \mathbf{w}^{\mathsf{H}}\mathbf{R}\mathbf{w}$$

Real-valued specialization:

$$\xi(\mathbf{w}) = \mathrm{E}[d^2(n)] - 2\mathbf{w}^{\mathsf{T}}\mathbf{p} + \mathbf{w}^{\mathsf{T}}\mathbf{R}\mathbf{w}$$

Gradient (real case):

$$\nabla_{\mathbf{w}}\xi = 2\mathbf{R}\mathbf{w} - 2\mathbf{p}$$

Wiener-Hopf / normal equations:

$$\mathbf{R}\mathbf{w}_o = \mathbf{p}, \qquad \mathbf{w}_o = \mathbf{R}^{-1}\mathbf{p}$$

Scalar form:

$$\sum_{i=0}^{M-1} w_{o,i}\, r_x(k-i) = p(k), \qquad k = 0,1,\ldots,M-1$$

Orthogonality principle:

$$\mathrm{E}[e_o(n)\mathbf{x}(n)] = \mathbf{0}$$

Minimum mean-squared error:

$$\xi_{\min} = \mathrm{E}[|d(n)|^2] - \mathbf{w}_o^{\mathsf{H}}\mathbf{p}$$

Using $\mathbf{R}\mathbf{w}_o = \mathbf{p}$,

$$\xi_{\min} = \mathrm{E}[|d(n)|^2] - \mathbf{w}_o^{\mathsf{H}}\mathbf{R}\mathbf{w}_o$$

7. Integral identity for optimum FIR $W_o(z)$

Equivalent contour form:

$$\xi_{\min} = \phi_{dd}(0) - \frac{1}{2\pi j}\oint_C W_o(z)\Phi_{xd}(z)\,\frac{\mathrm{d}z}{z}$$

Useful conversion identity:

$$\frac{1}{2\pi j}\oint_C z^{-i}\Phi_{xd}(z)\,\frac{\mathrm{d}z}{z} = \phi_{xd}(-i) = p_i$$

Hence the integral form and matrix form are equivalent:

$$\xi_{\min} = \mathrm{E}[|d(n)|^2] - \mathbf{w}_o^{\mathsf{H}}\mathbf{p}$$

8. Canonical form and eigenstructure

Complete-the-square form:

$$\xi(\mathbf{w}) = \xi_{\min} + (\mathbf{w} - \mathbf{w}_o)^{\mathsf{H}}\mathbf{R}(\mathbf{w} - \mathbf{w}_o)$$

For real-valued problems, let $\mathbf{v} = \mathbf{w} - \mathbf{w}_o$:

$$\xi = \xi_{\min} + \mathbf{v}^{\mathsf{T}}\mathbf{R}\mathbf{v}$$

If

$$\mathbf{R} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^{\mathsf{T}}, \qquad \boldsymbol{\Lambda} = \mathrm{diag}(\lambda_0,\ldots,\lambda_{M-1})$$

and $\mathbf{v}' = \mathbf{Q}^{\mathsf{T}}\mathbf{v}$, then

$$\xi = \xi_{\min} + \mathbf{v}'^{T}\boldsymbol{\Lambda}\mathbf{v}' = \xi_{\min} + \sum_i \lambda_i (v_i')^2$$

Contours are ellipses aligned with the eigenvectors of $\mathbf{R}$.

9. Converting a quadratic to matrix form

Map any quadratic to

$$\xi = \mathbf{w}^{\mathsf{T}}\mathbf{R}\mathbf{w} - 2\mathbf{w}^{\mathsf{T}}\mathbf{p} + \mathrm{E}[d^2(n)]$$

using: **Diagonal term** $c\,w_i^2 \Rightarrow R_{ii} = c$.
Cross-term $c\,w_i w_j \Rightarrow R_{ij} = R_{ji} = c/2$.
Linear term $\ell_i w_i \Rightarrow -2p_i = \ell_i$, so $p_i = -\ell_i/2$.

10. Noise-free channel equalization

If the channel noise is zero and the channel is $H(z)$, the zero-forcing equalizer is

$$W_o(z) = \frac{1}{H(z)}$$

Useful bilateral z-transform identity:

$$\frac{1}{1 - az^{-1}} \overset{\mathcal{Z}}{\longleftrightarrow} \begin{cases} a^n u[n], & |z| > |a| \\ -a^n u[-n-1], & |z| < |a| \end{cases}$$

For a stable inverse, choose an ROC that includes the unit circle.

11. Characteristic equations and eigenvalues

Always start from

$$\det(\lambda\mathbf{I} - \mathbf{R}) = 0$$

For

$$\mathbf{R} = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$$

$$\lambda^2 - 2a\lambda + a^2 - b^2 = 0, \qquad \lambda = a \pm b$$

For

$$\mathbf{R} = \begin{bmatrix} a & b & c \\ b & a & b \\ c & b & a \end{bmatrix}$$

$$\lambda^3 - 3a\lambda^2 + (3a^2 - 2b^2 - c^2)\lambda - a^3 + 2b^2(a-c) + ac^2 = 0$$

For real symmetric $\mathbf{R}$, eigenvalues are real and eigenvectors can be chosen orthonormal.

In $\mathbf{w}^{\mathsf{T}}\mathbf{R}\mathbf{w}$, the coefficient on $w_i w_j$ is $2R_{ij}$ when $R_{ij} = R_{ji}$.

If $v(n)$ is unit-variance white noise, then $\Phi_{vv}(z) = 1$ and $r_{vv}(k) = \delta(k)$.

12. Z-transform quick table

Convention: $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \quad q = z^{-1}$.

Common bilateral pairs

$x[n]$	$X(z)$	ROC / note
$\delta[n]$	1	all z
$\delta[n-k]$	z^{-k}	shifted impulse
$u[n]$	$\frac{1}{1-q}$	$\lvert z\rvert > 1$
$-u[-n-1]$	$\frac{1}{1-q}$	$\lvert z\rvert < 1$
$a^n u[n]$	$\frac{1}{1-aq}$	$\lvert z\rvert > \lvert a\rvert$
$-a^n u[-n-1]$	$\frac{1}{1-aq}$	$\lvert z\rvert < \lvert a\rvert$
$na^n u[n]$	$\frac{aq}{(1-aq)^2}$	$\lvert z\rvert > \lvert a\rvert$
$(n+1)a^n u[n]$	$\frac{1}{(1-aq)^2}$	$\lvert z\rvert > \lvert a\rvert$
$a^{n-k}u[n-k]$	$\frac{z^{-k}}{1-aq}$	$k \geq 0,\ \lvert z\rvert > \lvert a\rvert$
$u[n] - u[n-N]$	$\frac{1-z^{-N}}{1-q}$	finite, N samples
$a^n\{u[n] - u[n-N]\}$	$\frac{1-(aq)^N}{1-aq}$	finite length
$r^n\cos(\omega_0 n)u[n]$	$\frac{1 - r\cos\omega_0\, q}{D}$	$\lvert z\rvert > r$
$r^n\sin(\omega_0 n)u[n]$	$\frac{r\sin\omega_0\, q}{D}$	$\lvert z\rvert > r$

$$D = 1 - 2r\cos\omega_0\, q + r^2q^2$$

Two-sided nondecaying sinusoids/exponentials generally have no ordinary ROC.

Properties and ROC facts

Operation	Transform / note
Linearity	$\alpha x[n] + \beta y[n] \ \leftrightarrow\ \alpha X(z) + \beta Y(z)$
Delay/advance	$x[n-k] \ \leftrightarrow\ z^{-k}X(z)$
Time reversal	$x[-n] \ \leftrightarrow\ X(z^{-1})$; ROC inverted
Conjugate reverse	$x^*[-n] \ \leftrightarrow\ X^*(1/z^*)$
Exponential weight	$a^n x[n] \ \leftrightarrow\ X(z/a)$; ROC scaled by $\lvert a\rvert$
Convolution	$x[n] * h[n] \ \leftrightarrow\ X(z)H(z)$
Multiply by n	$nx[n] \ \leftrightarrow\ -z\frac{dX(z)}{dz}$
Right-sided rational	ROC outside outermost pole
Left-sided rational	ROC inside innermost pole
Stable LTI	unit circle in ROC
Causal stable rational	all poles strictly inside unit circle
Coefficient	$x[k] = \frac{1}{2\pi j}\oint X(z)z^{k-1}\,dz$